

Companion note

Variable relationships in the twin model

How each input moves each output — the shape of the curve, the direction, and where the leverage is. Read this beside the model explorer and the merged technical reference.

SCOPE

Qualitative. Shapes and directions, not derivations.

SOURCE

Equations 1–8, merged technical reference (SIH26120).

USE FOR

Shared intuition · slider reading · judge Q&A.

Revised 12 September 2026 to match the corrected technical reference: the cooling-constant law, the two-zone oil-rate law, the pump-capacity limit, the float-limit timing, the half-life of the temperature excess, and the Walther B spread. All numbers are for the worked case (2,000 m³, quality 0.70, 2.5 m stroke, 6 SPM, 120 production days).

One

The chain

The physics runs one way. Steam inputs on the left, the steam–oil ratio on the right. Nothing downstream feeds back upstream in this version.

steam volume + quality + temperature → **heated zone V_h , radius r_h** → cooling time constant τ
then, for every day of the cycle
 τ + day → **rock temperature $T(t)$** → **oil viscosity $\mu(t)$**
viscosity splits into two consequences
↓ mobility → **oil rate $q(t)$** , up to pump capacity | ↑ rod drag → **max safe pump speed $N_{\max}(t)$**
both meet at
SOR — barrels of steam per barrel of oil

The split at viscosity is the point to hold onto. Production and pump safety are not two problems that share a well. They are two results of one number, the reservoir temperature.

Two

How to read the shapes

Each relationship below is tagged with one of these shapes.

- **Linear** — output changes by the same proportion as the input. Double the input, double the output. A straight line.
- **Square root** — output grows, but slower and slower. Four times the input gives twice the output. Diminishing returns.
- **Inverse** — output is one divided by the input. The input goes up, the output comes down. Small inputs give very large outputs.
- **Exponential decay** — output falls fast at first, then ever more slowly, toward a floor. Set by a time constant.
- **Double-logarithmic** — steeper than exponential. A small input change produces a very large output change. This is the viscosity curve.
- **Saturating** — the relationship works up to a limit, then stops. More input past that point does nothing.
- **Integral** — the output is the running total of the input over time. It only rises; it rises fast when the input is high.

Three

Steam inputs → heated zone and duration

Steam volume V_s → heated-zone volume V_h **LINEAR**

Directly proportional. Twice the steam heats twice the rock volume, with steam temperature, quality and efficiency held fixed.

Equation 1.

Steam volume V_s → heated-zone radius r_h **SQUARE ROOT**

The radius grows as the square root of the volume. Four times the steam gives only twice the reach. Volume buys duration more than it buys distance.

$r_h = \sqrt{(V_h / \pi h)}$, after equation 1.

Steam volume V_s → cooling constant τ **SQUARE ROOT, THEN LEVELS OFF**

τ follows the heated zone's volume-to-surface ratio. A bigger zone has less cold surface for its size, so it holds heat longer. At first τ rises with the heated-zone radius, and so with the square root of steam volume. As the radius grows past the zone's thickness the gain shrinks, and τ levels off towards 55 days. There is no hard cap. Over 500–4,000 m³, τ runs from 21 to 35 days.

Equation 2: $\tau = 55.2 \cdot r_h / (r_h + h)$, with $h = 9.92$ m; $\tau = 30$ days in the worked case.

Steam quality x → heated zone and τ **NEAR-LINEAR**

Wetter steam carries less latent heat, so it heats less rock. The effect is real but small: across the whole 0.60–0.80 range the injected heat moves by about 17 percent. Same direction as steam volume, weaker.

Equation 1, enthalpy term $x \cdot L$.

Steam temperature T_s → day-zero rock temperature LINEAR

The temperature the well starts the cycle at moves one-for-one with the steam temperature. This sets how much headroom there is before viscosity climbs.

Equation 2, the $(T_s - T_i)$ term.

Steam temperature T_s → heated-zone volume and τ MILD INVERSE

Hotter steam spreads the same energy across a larger temperature gap, so the heated volume is slightly smaller and τ slightly shorter. Roughly a quarter smaller from 200 °C to 300 °C. It pulls against the day-zero benefit above, so raising T_s helps the early rate and slightly shortens the heat's life.

Equation 1, $(T_s - T_i)$ in both numerator and denominator.

Thermal efficiency η → heated zone and τ LINEAR

A straight scaling on the heat that actually reaches the rock. It is a calibration factor, not an operating decision, so the model holds it fixed at 0.5.

Equation 1.

Four

Duration and time → temperature

Day in the cycle t → rock temperature T EXPONENTIAL DECAY

Temperature falls fast in the first weeks, then slower, toward the native 50 °C. The excess over 50 °C halves every 0.69 τ days — about 21 days when $\tau = 30$.

Equation 2: $T(t) = T_i + (T_s - T_i) \cdot e^{-t/\tau}$.

Cooling constant τ → temperature on a given day SATURATING RISE

A larger τ stretches the cooling curve to the right, so on any given day the well is warmer. The gain per extra day of τ shrinks as τ grows, and τ itself levels off near 55 days.

Equation 2, through the $e^{-t/\tau}$ term.

Five

Temperature → viscosity

Rock temperature T → oil viscosity μ DOUBLE-LOGARITHMIC

This is the sharpest relationship in the model. As the oil cools, viscosity rises far faster than an exponential. From the worked figures: 250 °C gives 5 cP, 77 °C gives about 1,150 cP, 60 °C gives about 4,465 cP, and the last 10 °C down to 50 °C multiplies it by 2.6 to 11,500 cP. A small late-cycle temperature drop is a large thickening.

Equation 3, ASTM D341 / Walther: $\log_{10}(\log_{10}(\nu + 0.7)) = A - B \cdot \log_{10}(T)$. Do not approximate it with a plain exponential.

Walther slope B → viscosity μ PIVOT AT 50 °C

B is the temperature sensitivity of the oil. The curve is anchored at 50 °C, so every B gives the same viscosity there. Away from the anchor the curves for B = 3.2 and B = 3.8 separate. Every spread on this card is measured the same way: the higher of the two viscosities divided by the lower, minus one. The spread is 0% at 50 °C, 41% at 77 °C, 77% at 124 °C and 78% at 250 °C. Below the anchor it is 57% at 30 °C. Above 50 °C the higher B gives the thinner oil; below 50 °C it gives the thicker oil. The effect on the float limit is small: B from 3.2 to 3.8 moves the day the limit binds only from day 86 to day 91.

Equation 3. Open item: one reported viscosity range (10,000–13,000 cP at 50 °C), so B = 3.5 is assumed.

Six

Viscosity → the two consequences

Viscosity μ → max safe pump speed $N_{\max}$ INVERSE, THEN CAPPED

$N_{\max}$ is one divided by viscosity. Double the viscosity, halve the safe speed. For about the first three months (to day 88 at a 2.5 m stroke) the oil is thin enough that the 12 SPM mechanical ceiling (tier D) is the real limit and $N_{\max}$ is flat. Then viscosity blows up and the safe speed collapses within weeks.

Equation 5: $N_{\max} = K / (\mu \cdot S)$, then limited to 12 SPM.

Stroke length S → max safe pump speed $N_{\max}$ INVERSE

A longer stroke means the rod has to fall farther in the same half-stroke time, so the safe speed is lower. Going from a 1 m to a 3 m stroke cuts the viscosity-limited safe speed to a third, and brings the day the limit binds forward from day 169 to day 83.

Equation 5, S in the denominator.

Viscosity μ → oil rate q INVERSE, NATURALLY CAPPED

Thin oil flows more easily, so the rate rises as viscosity falls. The heated zone and the cold rock beyond it resist flow in series. Even when the heated oil is very thin, the cold zone still limits flow, so the rate has a natural ceiling — about 3.2 times the cold rate at a heated radius of 11.8 m. As the zone cools, the rate falls back to the cold rate of 10 bbl/d, not to zero.

Equation 4 (Boberg–Lantz, tier D): $q_{\text{in}} = q_{\text{cold}} \cdot \ln(r_e/r_w) / [(\mu_h/\mu_c) \cdot \ln(r_h/r_w) + \ln(r_e/r_h)]$.

Heated-zone radius r_h → peak oil rate RISES, DIMINISHING

A bigger heated zone moves more of the flow path into thin oil, so it raises the ceiling on the rate: 2.4 times the cold rate at 5.9 m, 3.2 times at 11.8 m, 3.9 times at 16.7 m.

Equation 4, the $\ln(r_e/r_h)$ term.

Pump capacity → oil rate q MINIMUM (BOTTLENECK)

The surface rate is the smaller of reservoir inflow and pump capacity at the set speed. Slowing the pump costs oil only when pump capacity falls below inflow. Late in the cycle inflow is 10–20 bbl/d, while the pump at 4.5 SPM can still move about 126 bbl/d, so slowing to the safe speed costs almost nothing.

Equation 4: $q = \min(q_{in}, A_p \cdot S \cdot N \cdot 1440 \cdot \eta_v)$.

*Seven***Everything → cumulative oil and SOR****Oil rate q over time → cumulative oil N_p** INTEGRAL

N_p is the running total of the daily rate. It climbs fastest early, when the rate is highest, then slows as the rate decays towards the cold rate. Late days still add oil at about the cold rate.

$N_p = \int q \, dt$ over the cycle.

Steam volume V_s → cumulative oil N_p CONCAVE

More steam keeps the well hot longer and raises its peak rate, so it produces more. But the gain shrinks with every extra cubic metre: the radius grows only as the square root of volume, and τ levels off. Over 120 days, 500 m³ gives 2,050 bbl, 2,000 m³ gives 2,950 bbl, and 4,000 m³ gives 3,620 bbl.

Chain through equations 1–4.

Steam volume V_s → SOR RISES

SOR is steam divided by oil. The steam on top rises linearly with V_s ; the oil underneath rises more slowly. So SOR rises with steam volume across the simulator range: 1.5 at 500 m³, 4.3 at 2,000 m³, 6.9 at 4,000 m³. Small volumes look efficient partly because SOR counts the cold oil the well would produce anyway.

$SOR = \text{steam} / N_p$.

Cycle length t_{cut} → SOR KEEPS FALLING, MORE SLOWLY

A longer cycle spreads the same steam over more oil, so SOR drops. The drop slows once the heat is gone, but the well still adds oil at about the cold rate, so SOR keeps falling: 6.8 at 60 days, 4.3 at 120, 3.0 at 240, 2.3 at 365. This model is single-cycle; it does not price the oil lost by not re-steaming sooner.

$N_p = \int q \, dt$, longer limit.

Stroke length S → SOR NONE UNLESS THE PUMP LIMITS

Stroke sets pump capacity and the safe speed. With the default settings pump capacity is far above inflow, so stroke does not change SOR. It matters only at low speed and short stroke, where the pump becomes the bottleneck: at 1 m and 2 SPM, SOR rises from 4.3 to 5.2. A longer stroke does bring the float limit earlier.

Through equation 4 (pump capacity) and equation 5.

Walther slope B → SOR MILD FALL

In the default case a higher B keeps the oil thinner through the warm part of the cycle, so the rate is a little higher and SOR is slightly lower: 4.35 at B = 3.2, 4.20 at B = 3.8.

Through equation 3 into equation 4.

Eight

Master table

Input → output	Shape	Direction	Strength
Steam volume → heated-zone volume	Linear	same ↑	Strong
Steam volume → heated-zone radius	Square root	same ↑	Medium
Steam volume → cooling constant τ	Square root, then levels off	same ↑ then flattens	Strong, then weak
Steam quality → heated zone, τ	Near-linear	same ↑	Weak (~17% span)
Steam temperature → day-zero temperature	Linear	same ↑	Strong
Steam temperature → heated-zone volume, τ	Mild inverse	opposite ↓	Weak
Thermal efficiency → heated zone, τ	Linear	same ↑	Medium (fixed in model)
Day in cycle → rock temperature	Exponential decay	falls ↓	—
Cooling constant τ → temperature that day	Saturating rise	same ↑	Strong early cycle
Rock temperature → oil viscosity	Double-logarithmic	$T \downarrow \Rightarrow \mu \uparrow \uparrow \uparrow$	Very strong
Walther B → viscosity (warm side)	Pivot at 50 °C	$B \uparrow \Rightarrow \mu \downarrow$ warm, $\uparrow$ cold	Moderate on μ , weak on float day
Viscosity → max safe pump speed	Inverse, then 12 SPM cap	$\mu \uparrow \Rightarrow N_{\max} \downarrow$	Very strong late cycle
Stroke length → max safe pump speed	Inverse	$S \uparrow \Rightarrow N_{\max} \downarrow$	Medium
Viscosity → oil rate	Inverse, naturally capped	$\mu \uparrow \Rightarrow q \downarrow$	Strong
Heated-zone radius → peak oil rate	Rises, diminishing	same ↑	Medium
Pump capacity → oil rate	Minimum (bottleneck)	only when capacity < inflow	Weak by default
Oil rate over time → cumulative oil	Integral	rises ↑, then slows	—
Steam volume → cumulative oil	Concave	same ↑, diminishing	Medium
Steam volume → SOR	Rises	worse ↑	Strong

Input → output	Shape	Direction	Strength
Cycle length → SOR	Keeps falling, more slowly	better ↓	Medium, then weak
Stroke length → SOR	None unless the pump limits	—	None by default
Walther B → SOR	Mild fall	B ↑ ⇒ SOR ↓	Weak

Nine

Three non-linearities to remember

1 · Viscosity versus temperature is not gentle

It is double-logarithmic — steeper than an exponential. A 10 °C drop near reservoir temperature can double or triple the viscosity. This is why the last weeks of a cycle are dangerous and why the model refuses a simple exponential fit.

2 · Steam volume has diminishing returns

Reach grows as the square root of volume, and the cooling constant levels off near 55 days. Each extra cubic metre adds less oil than the one before, while it adds fully to the numerator of SOR.

3 · The lowest SOR is not the most oil

SOR is lowest at small steam volumes, but those settings also produce the least oil. The optimisation is a trade-off, not a single dial. Rod safety enters the same way: running the pump slow forever avoids every failure and produces almost nothing.

Ten

What this model does not capture

- **Reservoir pressure is held constant.** It drives inflow directly but has no depletion or support term yet. The reported 116 ksc is itself uncertain (technical reference, open item 3).
- **The crude is non-Newtonian.** Viscosity depends on shear rate as well as temperature, so pump speed feeds back into the safe pump speed. That loop is not built.
- **Single cycle only.** The cycle-length relationship does not price the oil lost by re-steaming too late.
- **The oil-rate and SOR curves are an estimate.** The cold rate of 10 bbl/d is set so the worked example (SOR 4.27) sits inside the reported 3.0–5.2 band; it is not measured. Treat the shapes as sound and the absolute numbers as indicative.
- **The pump-speed limit is a static force balance.** It ignores rod stretch and fluid load, uses the heated-zone viscosity for the colder oil in the tubing, and compares fall speed with the mean rather than the

peak rod speed. So it reads optimistic and needs calibration against failure history.

Baghewala Digital Twin · SIH26120 · Companion to the merged technical reference. Equations and parameter ranges: sections 4–5 of that document. Model source: `src/lib/physics/`, checked against these documents with `npm run verify:physics`. Explore the curves at `/simulator` and `/research`.